MedDiag

AI-Based Medical Diagnosis Assistant

ECG / EEG / Chest X-ray analysis with explainable AI and automated reporting

Academic Project Technical Report
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Project purpose. MedDiag is a research and educational prototype that accepts biomedical signals or a chest radiograph, applies preprocessing and a modality-specific model, extracts interpretable clinical features, and returns a structured decision-support response with visual explanations and report files.

Important limitation. The output is not a medical diagnosis or a medical device. It must not be used for clinical decisions. Results in this report are engineering and research evaluation results, not evidence of clinical safety.

1. Executive Summary

MedDiag combines a FastAPI backend, a dependency-free web dashboard, three neural inference pipelines, clinical feature extraction, explainable-AI visualisations, SQLite audit storage, and PDF/HTML/JSON report generation. The same API supports ECG and EEG numeric samples and uploaded chest X-ray images.

At startup, the application creates required directories and the audit database, loads the ECG, EEG and X-ray models through a central registry, and exposes health, analysis, history, model, demo and report routes. If a checkpoint is unavailable or PyTorch cannot be used, a deterministic heuristic baseline is selected and the response explicitly reports model_source = heuristic_baseline.

Main objectives

- Detect five ECG classes: NORM, AFIB, MI, STDE and PVC.
- Detect four EEG classes: NORMAL, EPILEPSY, PARKINSON and DEPRESSION.
- Detect six chest X-ray classes: NORMAL, PNEUMONIA, CONSOLIDATION, EFFUSION, TUBERCULOSIS and COVID19.
- Attach measurements, quality metrics, recommendations and explainability to every analysis.
- Provide reproducible tests, benchmarks and downloadable reports.

2. What the Repository Contains

Path	Role
app/main.py	Creates the FastAPI application, lifespan, middleware, dashboard and documentation routes.
app/api/	REST endpoints for ECG, EEG, X-ray, health, models, history, demo and reports.
app/schemas.py	Pydantic request and response contracts, validation and output fields.
app/ml/ecg/	ECG preprocessing, clinical features, ResNet1D model and inference.
app/ml/eeg/	EEG preprocessing, spectral features, EEGNet model and inference.
app/ml/xray/	Image preprocessing, radiograph features, CNN model and inference.
app/ml/xai/	Grad-CAM, integrated gradients, occlusion and visualisation helpers.
app/services/diagnosis.py	Orchestrates inference, findings, recommendations and XAI.
app/services/report.py	Renders analysis payloads as PDF, HTML or JSON.
app/core/	Logging, audit SQLite database, storage, parsers, errors and metrics.
app/web/static/	Offline dashboard HTML, CSS, JavaScript and SVG assets.
scripts/	Server, training, dataset, evaluation, benchmark, smoke and verification tools.
tests/	API, feature, explainability and data-splitting unit/integration tests.
models/	Trained model checkpoints and recorded metrics.
results/	Benchmark summary, per-class metrics and confusion matrices.
reports/ and storage/	Generated reports, analysis payloads, logs and audit data.

3. System Architecture and Working

The end-to-end processing sequence is:

1. **Receive input.** The API accepts a validated JSON body or multipart upload.

2. **Validate.** Pydantic checks patient reference, sampling rate, sample count, image type and field limits.

3. **Parse and preprocess.** Signals are converted to numeric arrays; images are decoded, checked and normalised.

4. **Extract features.** Domain measurements are computed, such as heart-rate/HRV and morphology for ECG, spectral/Hjorth features for EEG, and quality/zone statistics for X-ray.

5. **Run the modality model.** The registry selects ECG ResNet1D, EEGNet or the four-stage XrayCNN. The model produces class scores, which are converted into class probabilities.

6. **Apply quality and interpretation logic.** Signal quality, findings, severity, differential classes and recommendations are assembled.

7. **Generate explanations.** Depending on the path, Grad-CAM, integrated gradients, occlusion sensitivity and feature attribution produce visual or ranked evidence.

8. **Persist and respond.** The complete payload is stored, recorded in the audit trail, returned as JSON and can be rendered as PDF, HTML or JSON.

Model and feature summary

Modality	Input representation	Model	Fused clinical features	Output classes
ECG	One-dimensional time series; resampled to 500 Hz; 5-second / 2,500-sample window	ResNet1D + feature-fusion head	18	5
EEG	One-dimensional time series; typically 256 Hz; fixed model window	EEGNet + feature-fusion head	16	4
Chest X-ray	Greyscale image resized to 224 × 224 and normalised	Four-stage CNN + feature-fusion head	11	6

The application reports whether the result came from a trained checkpoint or the heuristic baseline. This prevents a fallback rule-engine result from being mistaken for a learned-model result.

4. Inputs and Outputs

ECG and EEG input

The JSON signal request contains a de-identified patient object, a flat list of time-ordered floating-point samples, and sampling_rate in Hz. Optional fields include lead_names, channel, window_seconds and context. Upload endpoints accept CSV, JSON, whitespace-separated TXT or MAT files.

Chest X-ray input

The X-ray endpoint receives a PNG, JPEG or DICOM file as multipart form data, plus patient_id, optional age and sex, and an acquisition view such as PA or AP.

Response output

Output field	Meaning
analysis_id	Unique identifier used for history and report retrieval.
prediction	Predicted label, confidence, severity, description and all class probabilities.
differential	Ranked alternative classes and probabilities.

Output field	Meaning
findings	Named measurements, values, units, reference ranges, status and explanations.
quality	Signal-quality index, usability, SNR, clipped samples and quality issues.
explainability	Methods, top contributing features, relevance and base64 PNG images.
recommendations	Follow-up or review suggestions generated by the service.
model_source	trained or heuristic_baseline; essential context for interpreting results.
disclaimer	Safety notice included in every diagnosis response.

5. How a Result Is Produced

ECG processing

The ECG path auto-detects units, resamples to 500 Hz, removes baseline wander, applies a 0.05–40 Hz band-pass and a 50 Hz notch where appropriate, selects a high-energy five-second window and z-score normalises it. Pan-Tompkins-style processing identifies R-peaks. RR intervals, heart rate, HRV, QRS and ST measurements, and morphology-based PVC evidence become clinical features for the model and report.

EEG processing

The EEG path normalises and windows the signal, then calculates band powers for delta, theta, alpha, beta and gamma, spectral entropy, Hjorth parameters, spike-wave index, alpha asymmetry and artefact indicators. These measurements are fused with the EEGNet representation before the class probabilities are produced.

Chest X-ray processing

The image path decodes the uploaded image, checks whether it is plausible as a chest radiograph, applies enhancement and normalisation, and resizes it to 224 × 224. Image-quality and lung-zone statistics are combined with the CNN representation.

Interpretation and explainability

The highest-probability class becomes the primary prediction. The probability vector is retained so alternatives are visible. Quality metrics can reduce confidence or mark an input unusable. XAI methods expose which time segments, pixels or clinical features influenced the score; the visualisations are embedded as base64 PNG data in the response and reports.

Report generation

POST /api/v1/reports/{analysis_id}?format=pdf|html|json loads the persisted analysis payload. ReportLab creates the PDF, a standalone template creates HTML, and JSON preserves the raw payload. The report is saved under reports/ and registered in the audit trail for later download.

6. Testing and Reproducibility

The project separates software correctness tests from model evaluation. This is important because a passing API test proves that the pipeline works, while a benchmark metric describes performance on a defined dataset and split.

Test area	What is checked
API and integration	Health, model registry, ECG/EEG JSON and upload paths, X-ray analysis, validation errors, report generation/download, history, statistics, demo routes and dashboard assets.
Feature extraction	Filtering, resampling, z-scoring, R-peak detection, preprocessing shapes, quality metrics, class separation and reference tables.
Explainable AI	Plotting helpers, ECG/X-ray Grad-CAM, integrated gradients, feature attribution and occlusion sensitivity.
Data splitting	Disjoint grouped splits, completeness, zero group overlap, determinism, ratio budgets and macro-F1 handling.

Reproduction commands

```
.venv\Scripts\python.exe -m pytest -q
.venv\Scripts\python.exe scripts\smoke_check.py
.venv\Scripts\python.exe scripts\evaluate_models.py
.venv\Scripts\python.exe scripts\run_benchmark.py --suite full --per-class 40 --validation-ratio 0.15
--test-ratio 0.15 --out results
```

The documented automated suite contains 54 unit/integration tests. Benchmark evaluation uses a global seed of 42 for training/splitting and seed 90210 for synthetic evaluation. Real benchmark splits are grouped by record or

subject to avoid leakage between train, validation and test partitions.

7. Results

The results below are taken from results/summary.csv. The real-data rows are the appropriate rows to discuss as external evaluation; synthetic rows demonstrate pipeline behaviour in a deliberately controlled domain and must not be read as clinical accuracy.

Evaluation set	Modality	n	Accuracy	Macro-F1	Macro-AUC	Mean ms
mit-bih	ECG	145	0.5931	0.5770	0.7289	40.8
bonn	EEG	80	0.8875	0.8861	0.9559	8.4

Interpretation

On the untouched real-data test partitions, the real-benchmark ECG model achieved 0.5931 accuracy, 0.5770 macro-F1 and 0.7289 macro-AUC on 145 MIT-BIH windows. The real-benchmark EEG model achieved 0.8875 accuracy, 0.8861 macro-F1 and 0.9559 macro-AUC on 80 Bonn recordings. Both exceeded their heuristic baselines.

The trained synthetic-domain models reached 1.0000 on held-out synthetic sets. This is expected because the generator encodes separable signatures; it confirms the end-to-end pipeline and does not establish performance on real patients. The lower transfer scores also show why dataset provenance and domain matching matter.

8. Limitations, Safety and Future Work

- Synthetic data are not a substitute for clinical validation or representative patient cohorts.
- The ECG real-data evaluation is modest and has a small grouped test partition; AFIB is a rhythm-level property that is difficult to infer from a short beat-centred window.
- Model confidence is not clinical certainty. Users must review raw input, quality metrics, findings and alternatives.
- Future work includes broader external validation, larger subject-independent cohorts, calibration, prospective review, fairness analysis and formal clinical safety assessment.

9. Conclusion

MedDiag is a complete academic software prototype: it connects validated input handling, modality-specific preprocessing, neural inference, interpretable features, explainable visualisation, persistence, testing and report generation in one offline service. Its strongest contribution is the traceable end-to-end workflow and explicit separation between trained-model results, heuristic fallback results, synthetic evaluation and real-data evaluation.

Primary project references: README.md for setup and usage; TESTING.md for parameters, benchmark protocol and detailed results; requirements.txt for dependencies; tests/ for executable verification.